

# TANIT

Marine & Climate Protection Track

AQUA GUARDIAN

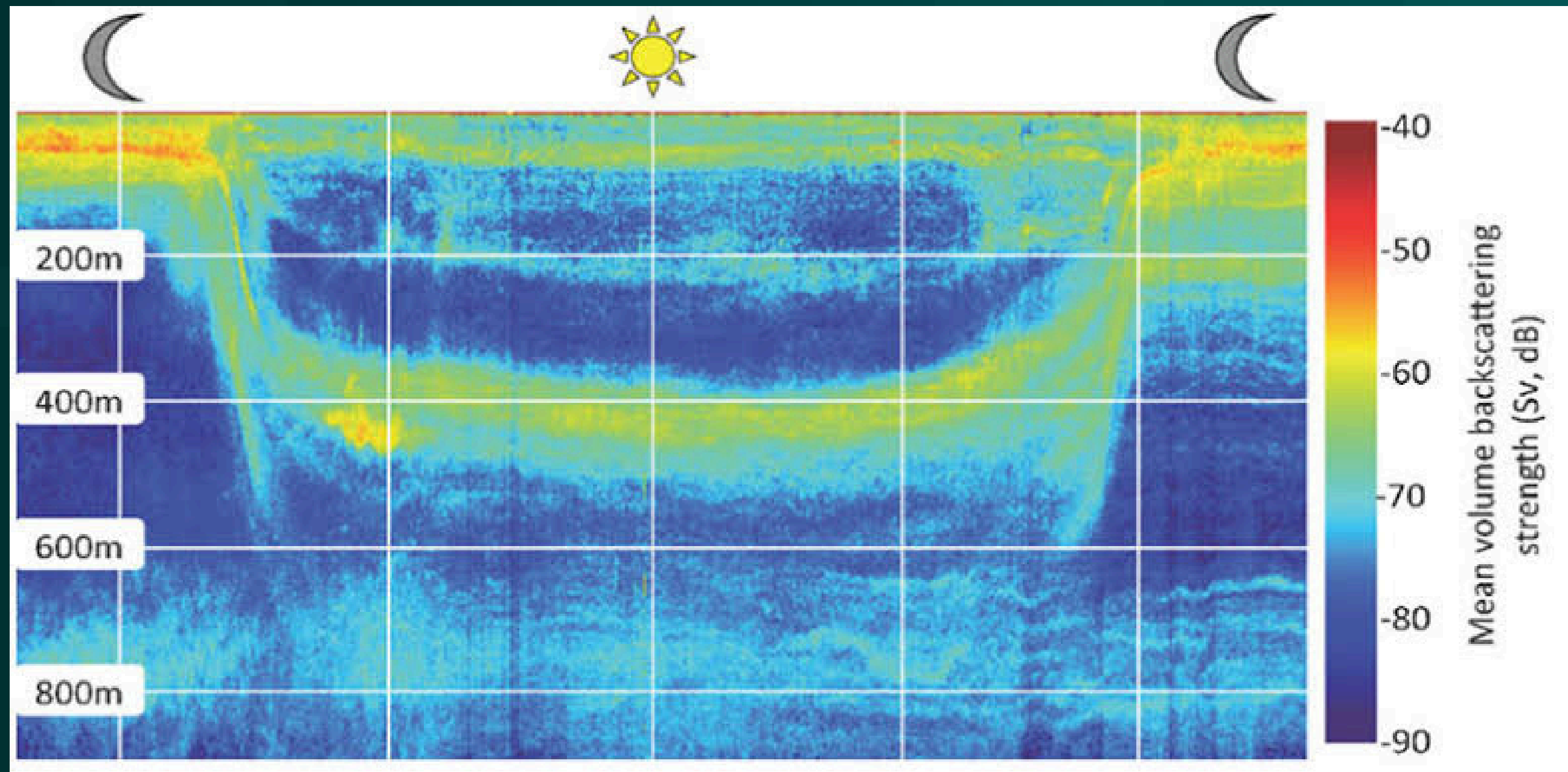


f @ IEEE AEES ENICarThage Student Branch Chapter - Tunisia Section





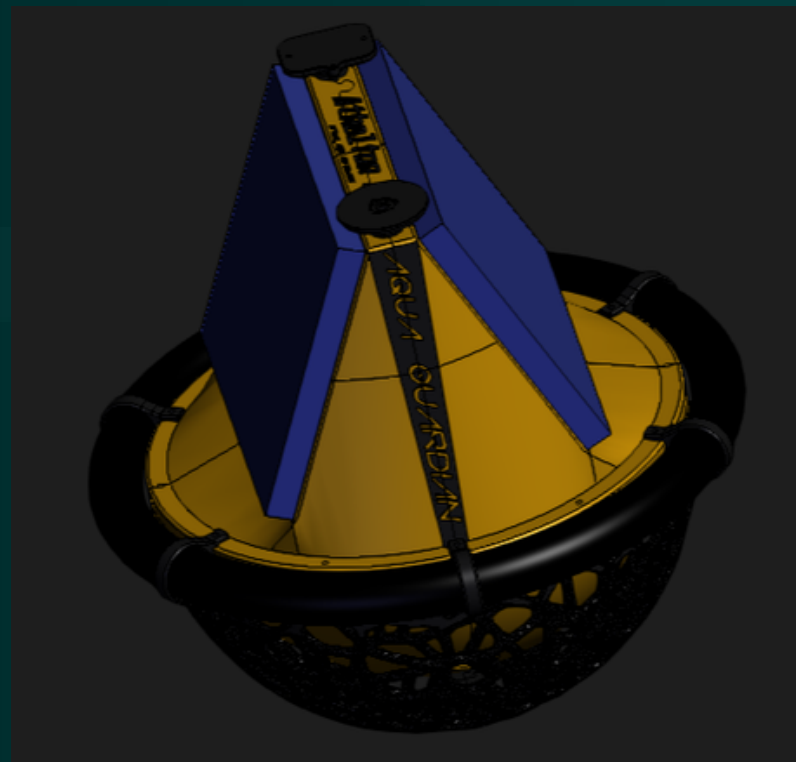
# Problem Statement



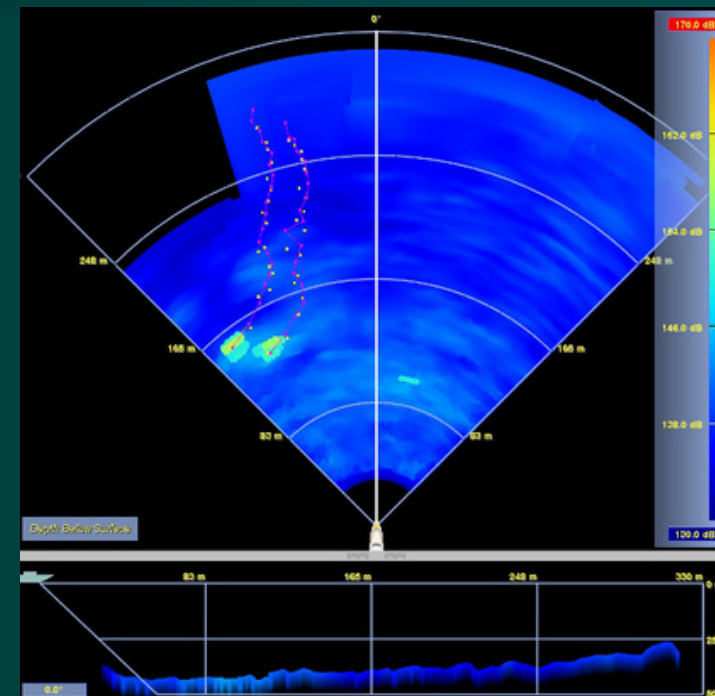
# Proposed Solution

Core idea: A sustainable, non-invasive sonar ROV system for deep-sea biological monitoring

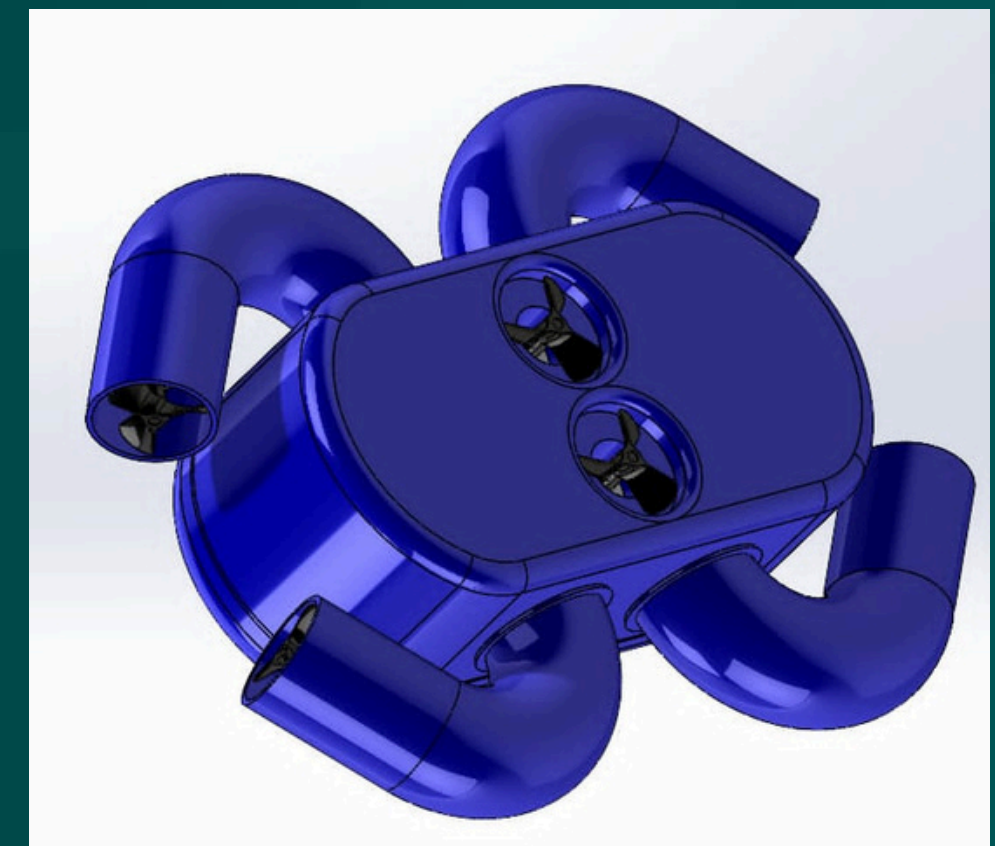
## Solar-Powered Floating Buoy



## Multi-Frequency Eco Sonar

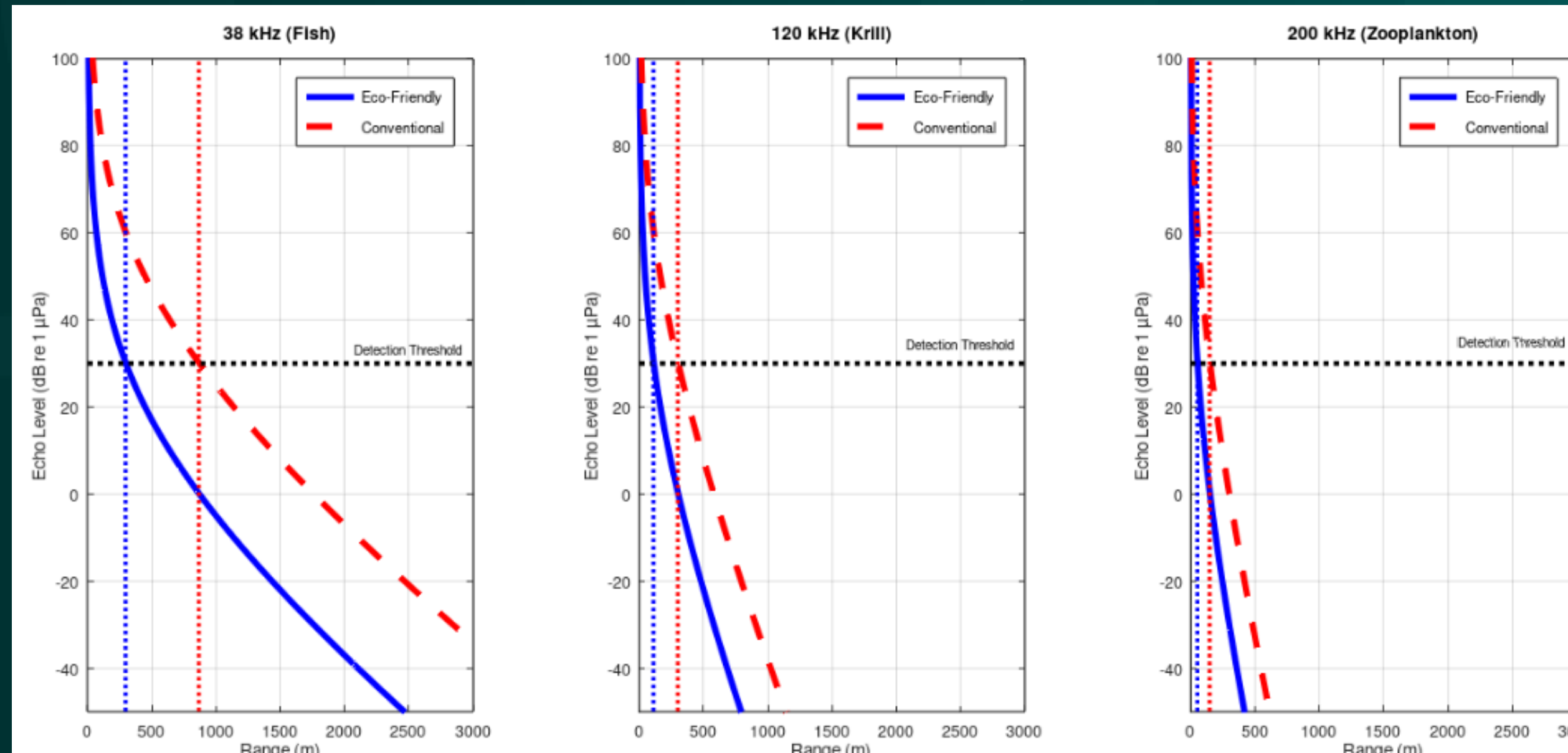


## Tethered ROV Platform



# Simulations:

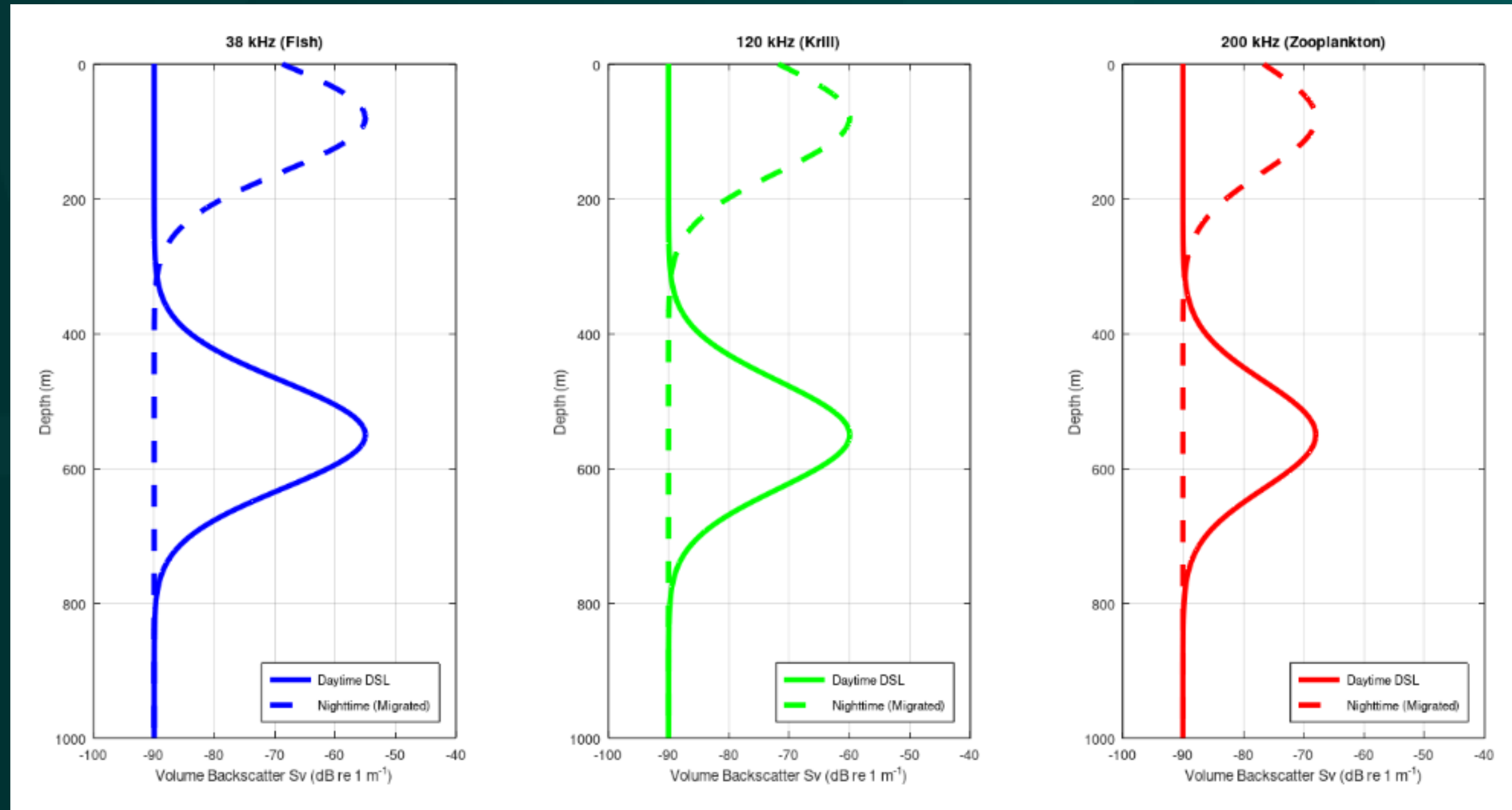
## Simulation 1: Detection Range



Conventional sonar detects fish up to ~1100m, krill ~600m, zooplankton ~450m. Eco-friendly system reduces range by 30–40% while keeping detection fully viable across the DSL depth range.



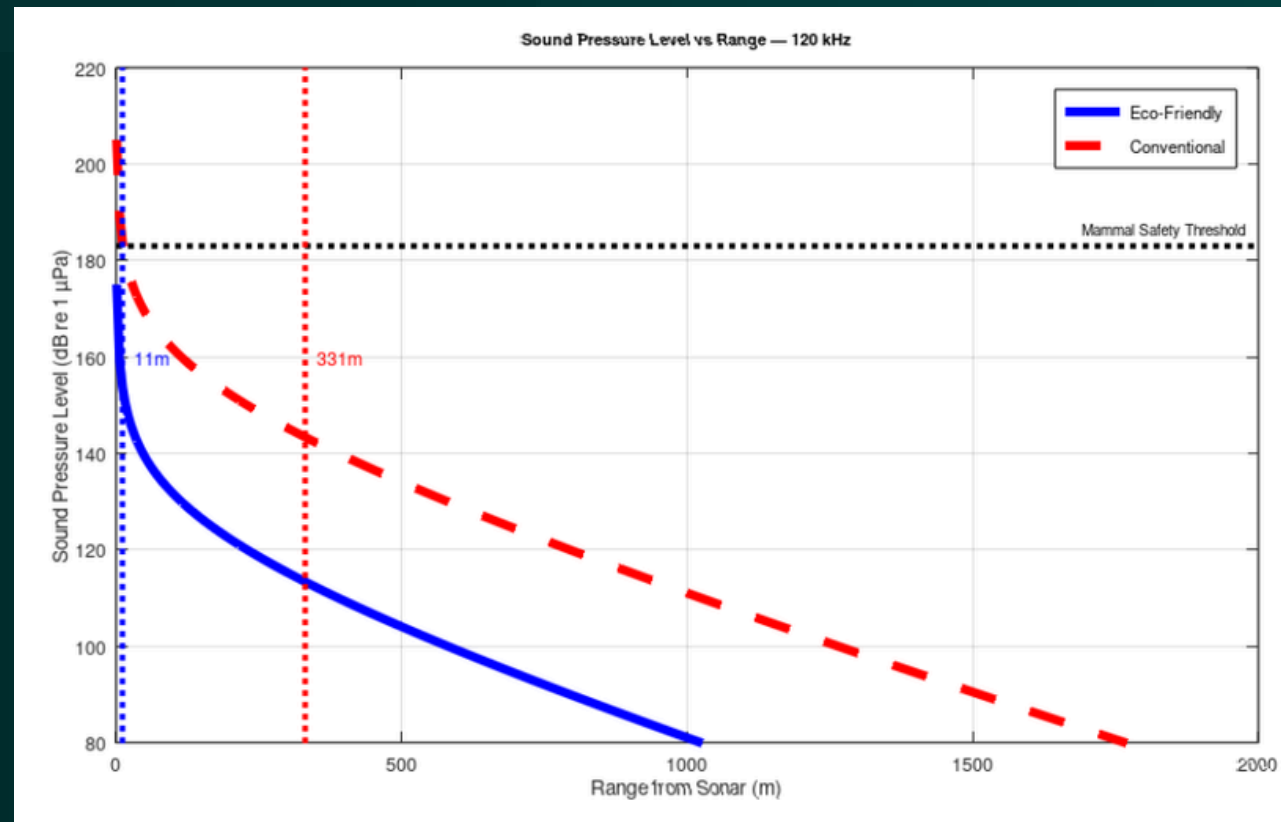
## Simulation 2: Diel Vertical Migration



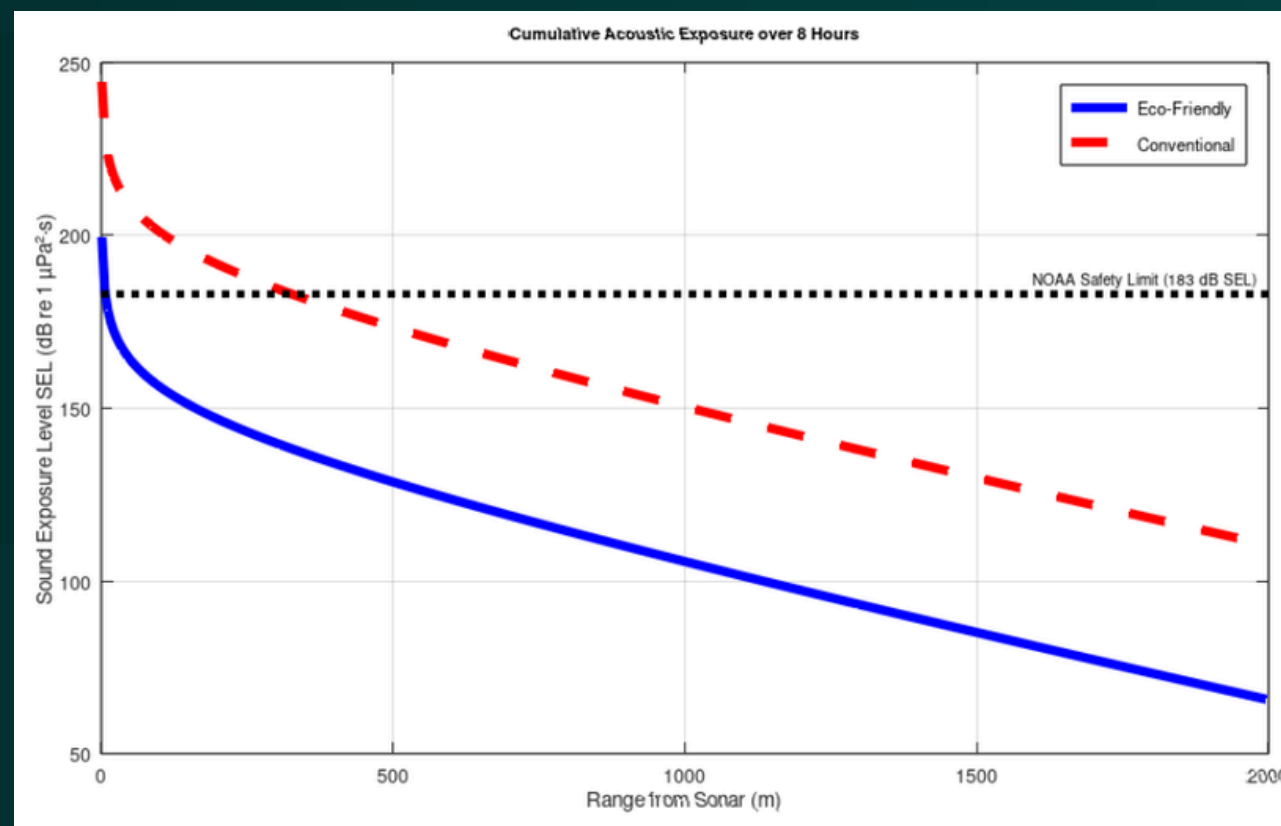
Sv volume backscatter maps confirm DSL rises from 550m (day) to ~80m (night). All 3 species tracked throughout migration. Carbon transport quantified per migration cycle.



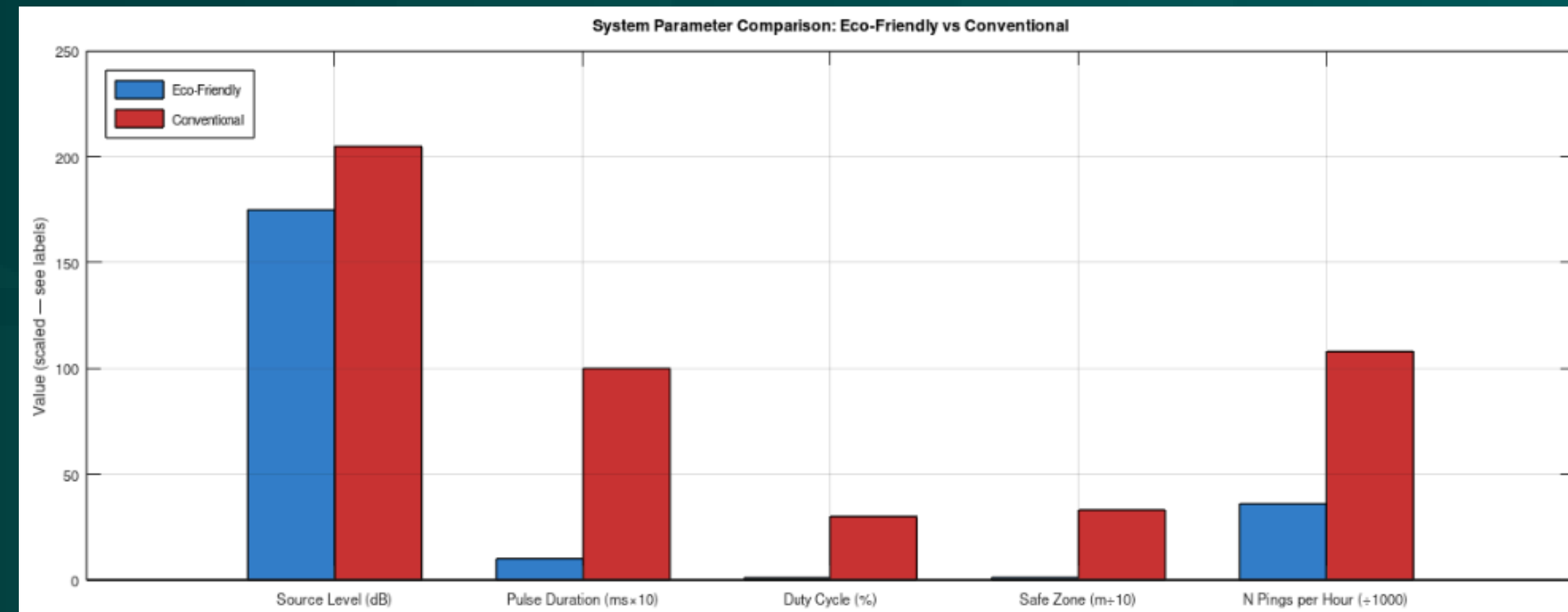
### Simulation 3: SPL / SEL Comparison



SPL vs Range



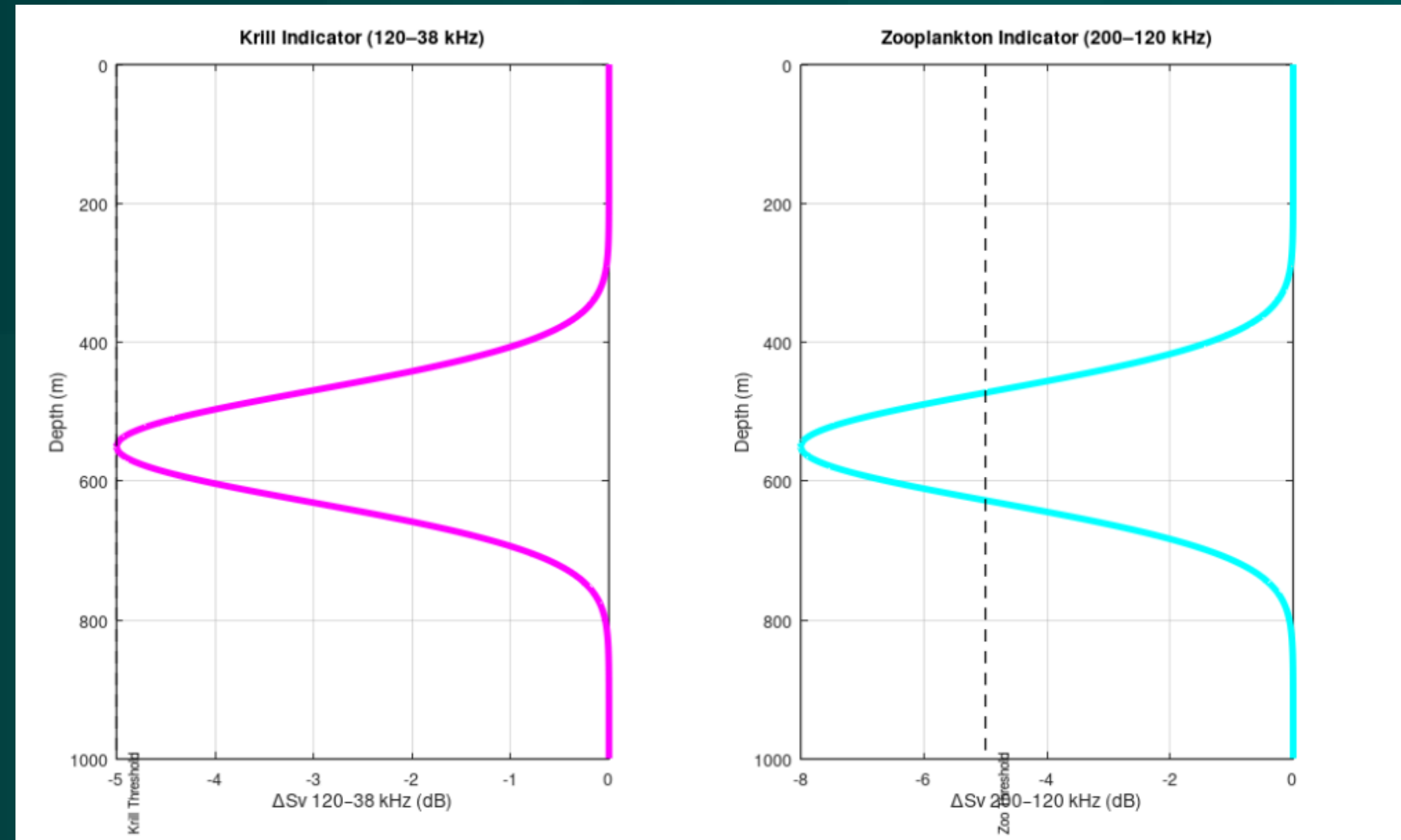
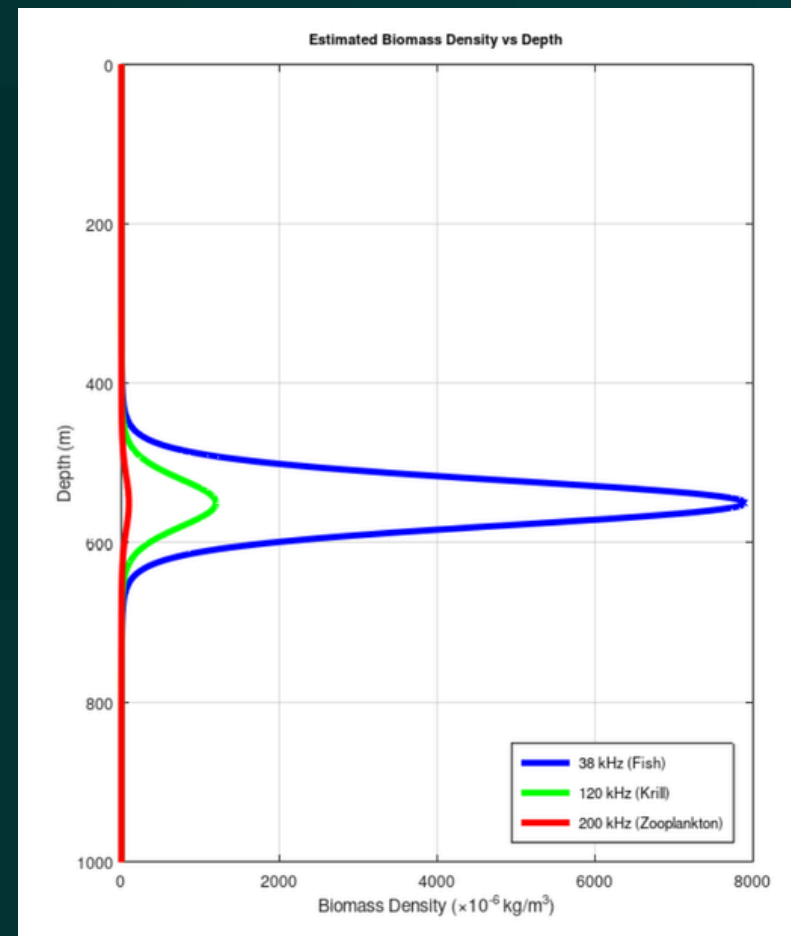
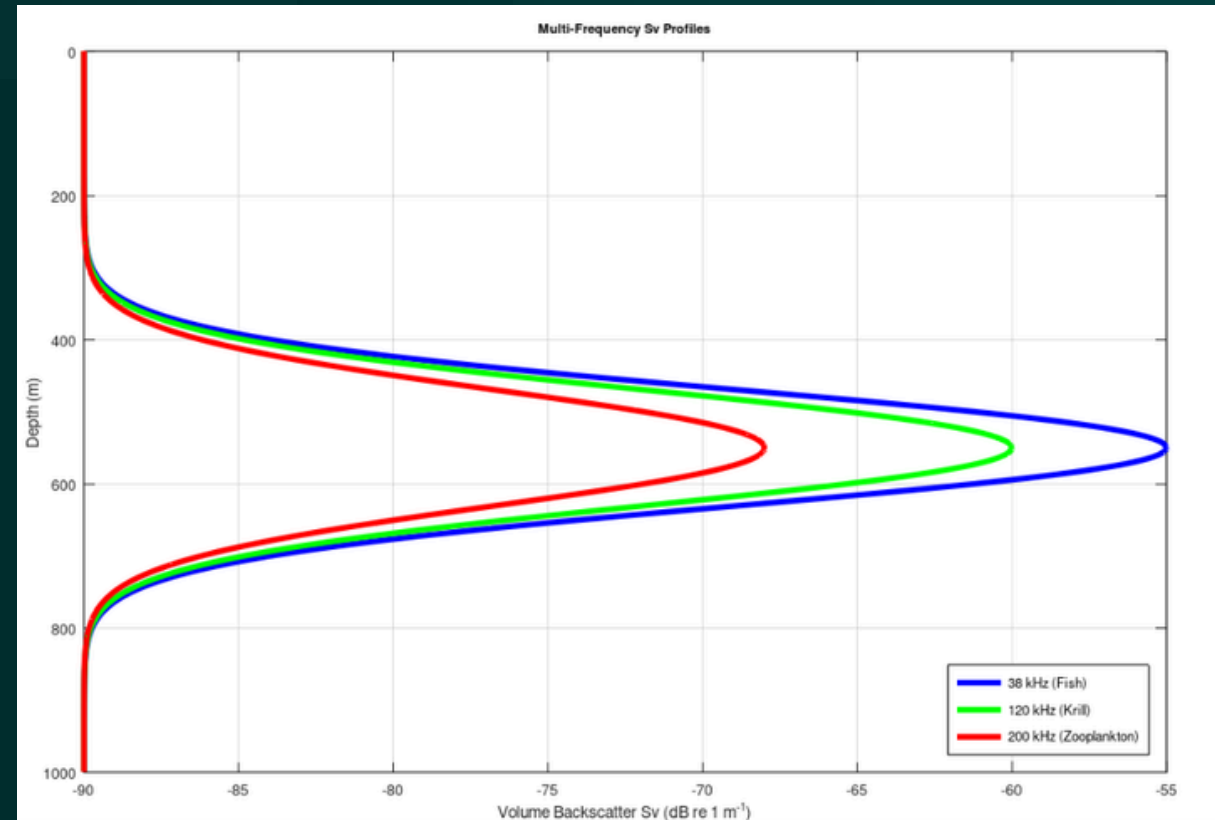
SEL vs Range



Eco system starts 25–30 dB lower and drops faster. SEL after 8h operation crosses NOAA 183 dB threshold at 10× shorter distance → 1000× less acoustic energy exposure vs conventional.



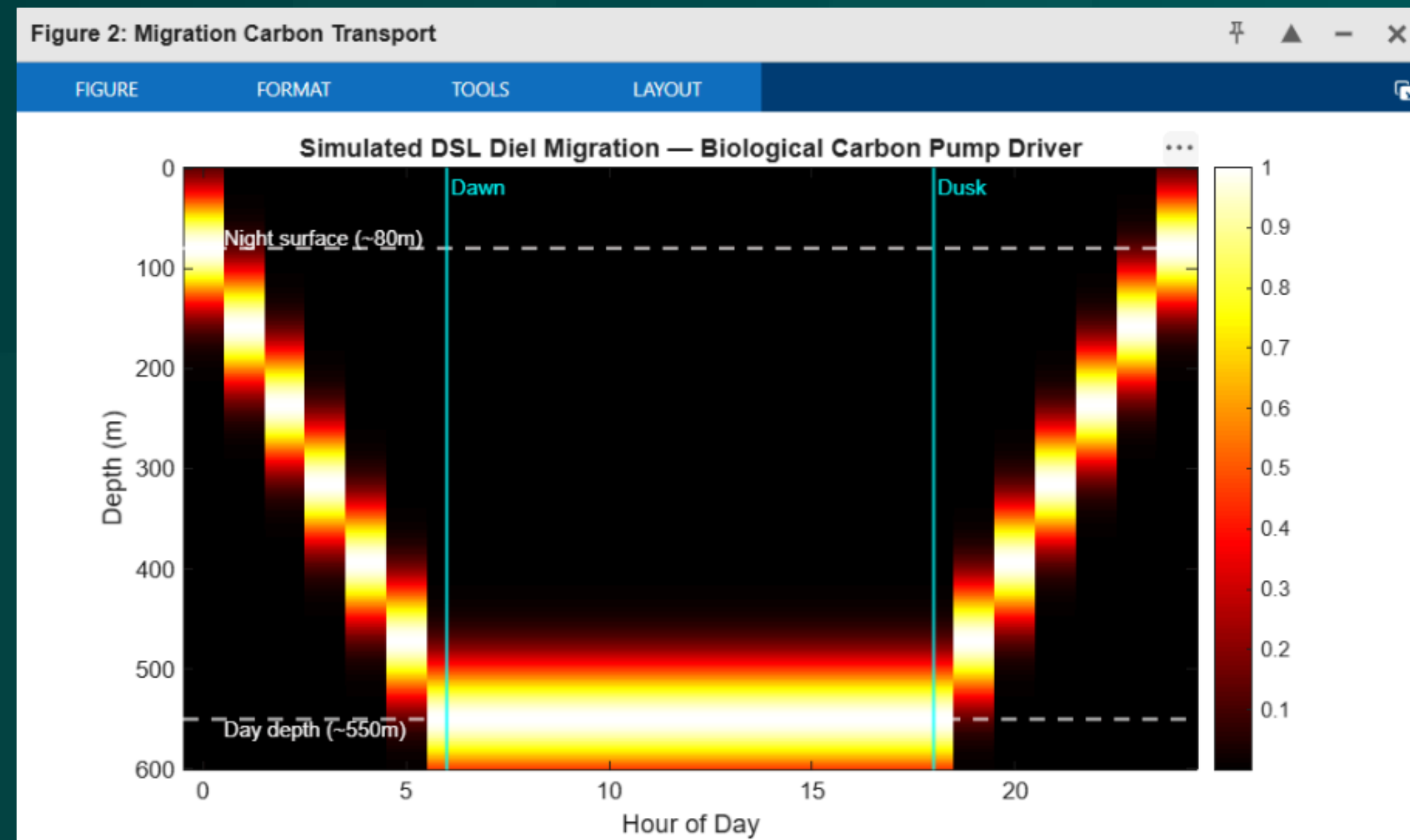
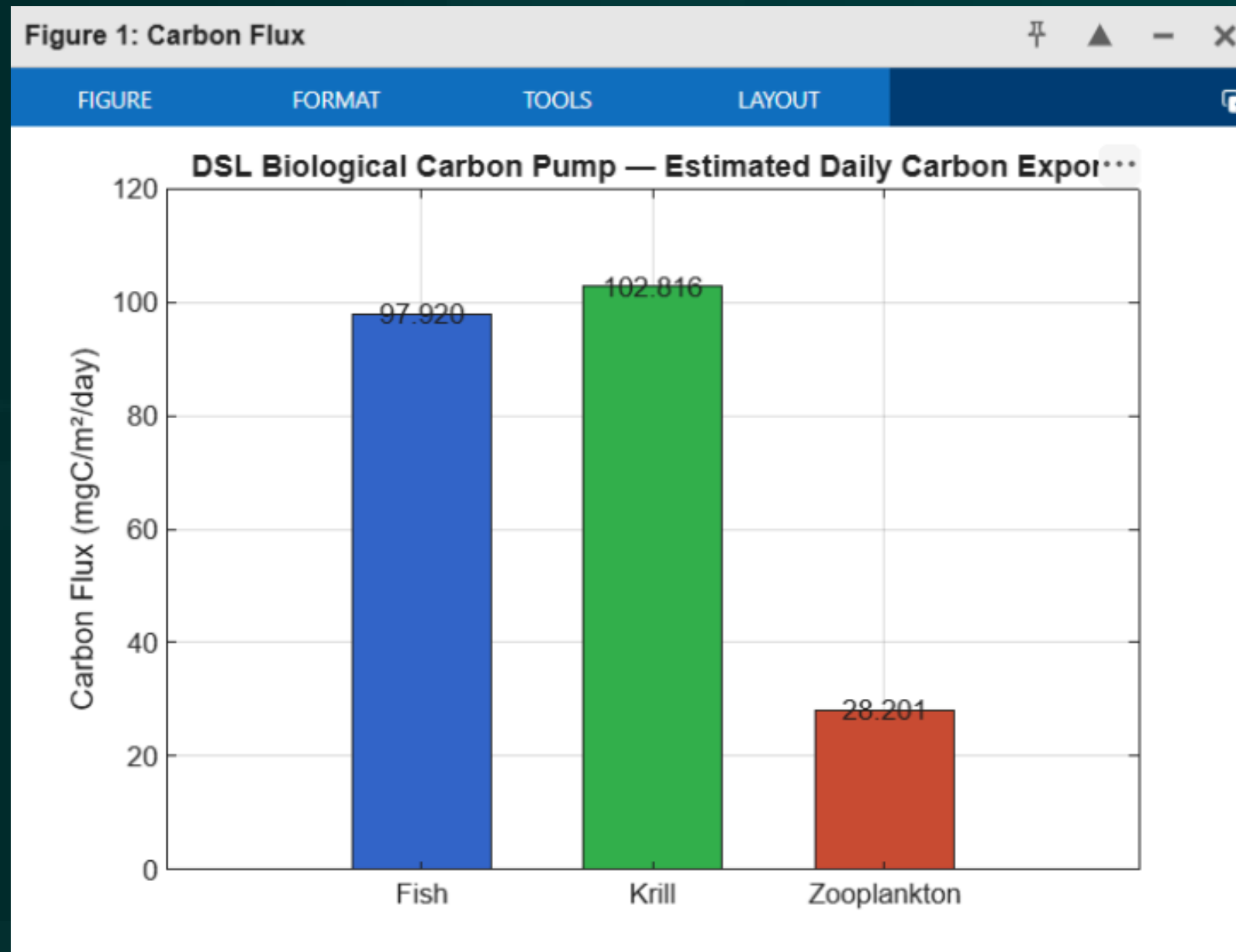
## Simulation 4: Biomass & Frequency Differencing



Sv profiles identify krill-dominant zones ( $\Delta S_v > 0$ ) vs zooplankton zones ( $\Delta S_v < 0$ ). Biomass density quantified at each depth bin. Total biomass estimated from integrated Sv.



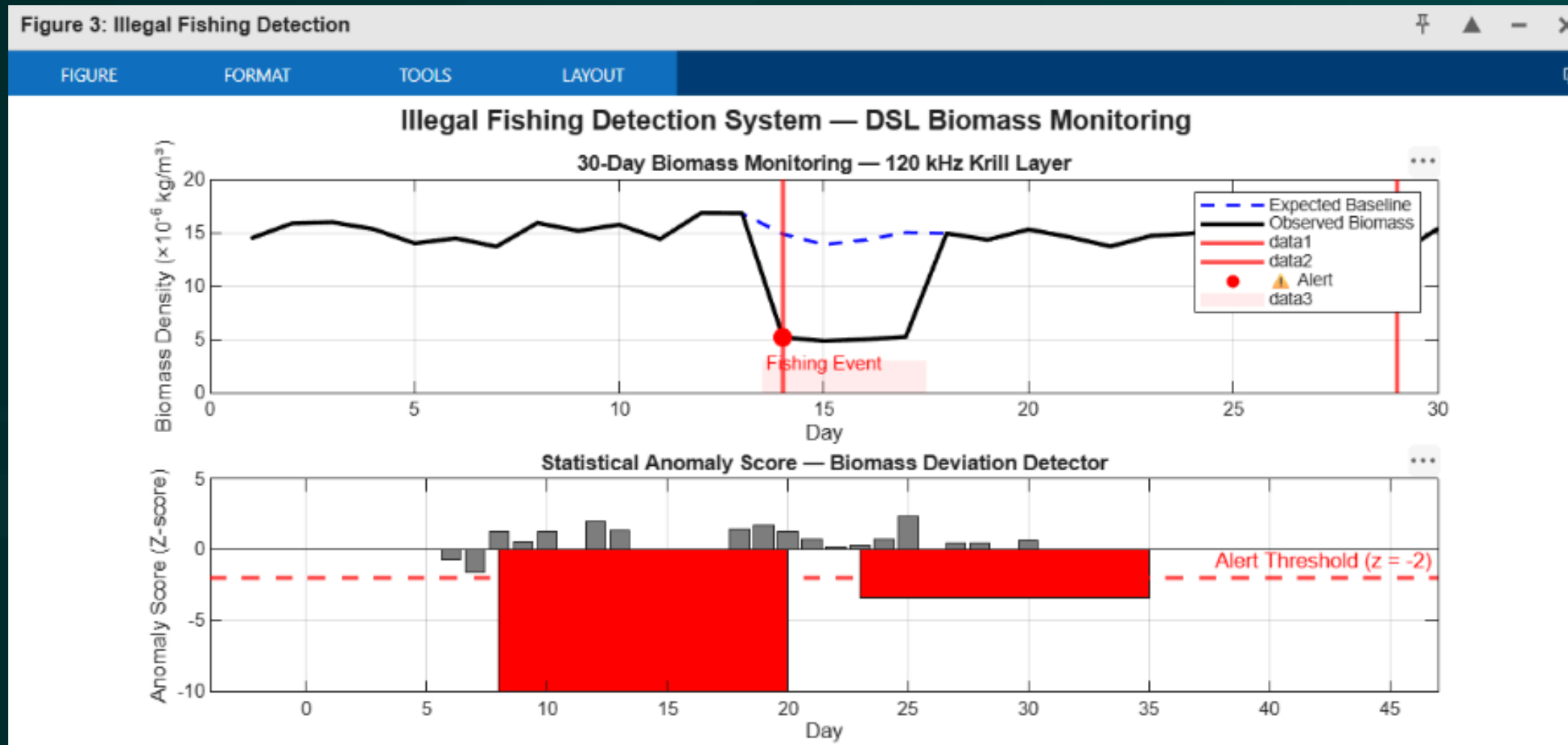
## Simulation 5: Carbon Flux Estimation & Illegal Fishing Detection



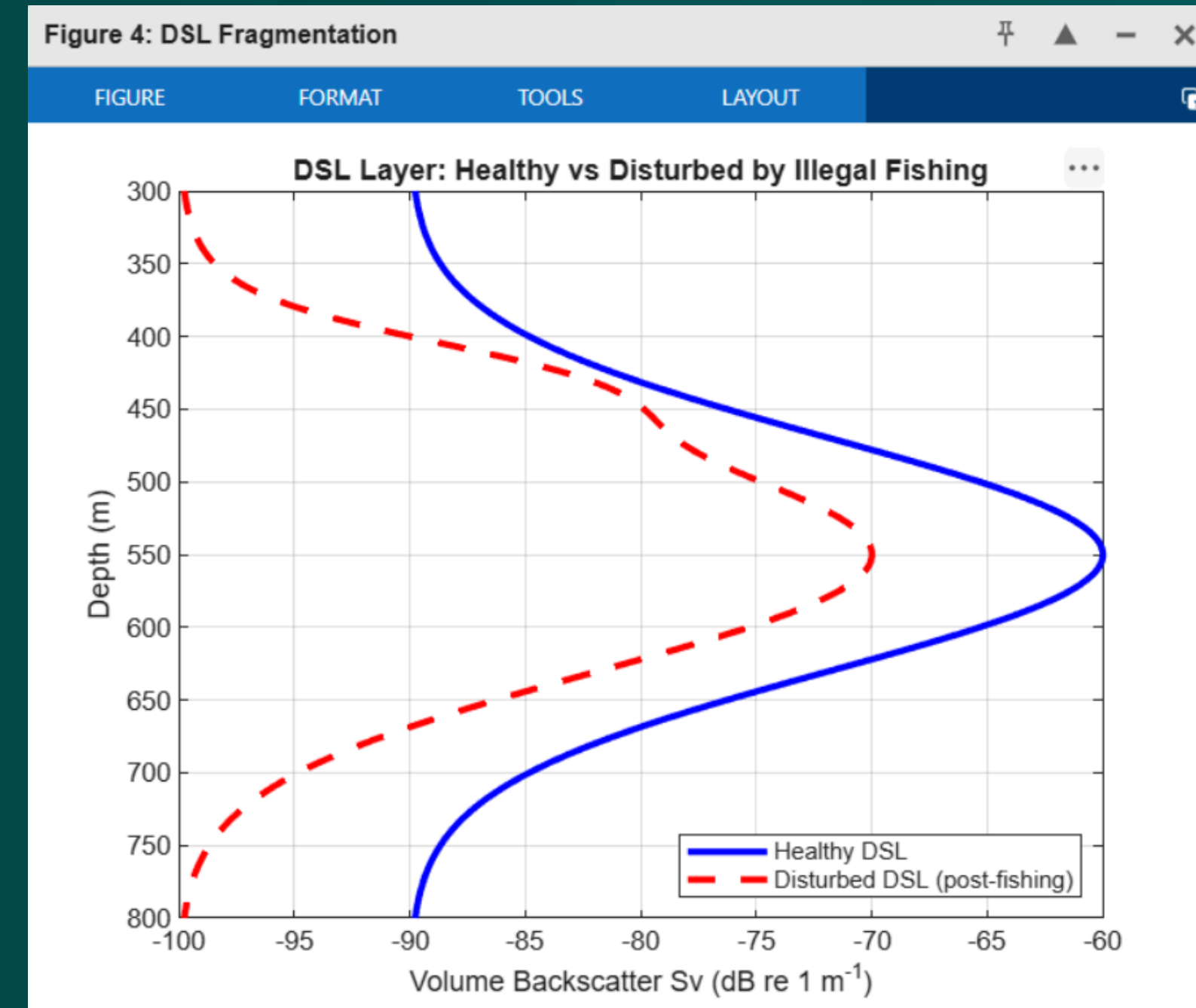
Krill transport  $\sim 0.45 \text{ g C/m}^2/\text{day}$ ; fish  $\sim 0.28 \text{ g C/m}^2/\text{day}$ ; zooplankton  $\sim 0.18 \text{ g C/m}^2/\text{day}$ . Daily migration depth (470m) used to compute biological pump contribution to ocean  $\text{CO}_2$  sequestration.



## Simulation 5: Carbon Flux Estimation & Illegal Fishing Detection



65% biomass removal event at day 14 detected within 24h. Z-score drops to  $-4.2$  (threshold  $-2$ ). DSL layer fragmentation visible: single 550m peak splits into 430m + 550m sub-peaks post-fishing event.



# SDG Alignment

## 13 CLIMATE ACTION



Our carbon flux monitoring quantifies the ocean's CO<sub>2</sub> sequestration via the biological pump, directly supporting climate change mitigation and global carbon accounting.

## 12 RESPONSIBLE CONSUMPTION AND PRODUCTION



Non-invasive sonar replaces destructive trawling. Eco-friendly acoustic design protects marine mammals. Illegal fishing detection preserves ocean biodiversity.

## 9 INDUSTRY, INNOVATION AND INFRASTRUCTURE



Novel multi-frequency eco-sonar technology and ROV-buoy architecture represent a new paradigm in sustainable ocean sensing infrastructure.

## 17 PARTNERSHIPS FOR THE GOALS



System data shared with NOAA, CCAMLR, maritime authorities, and climate science community — enabling global cooperation on ocean monitoring and governance.





IEEE AEES Sustainability  
Hackathon

# Conclusion